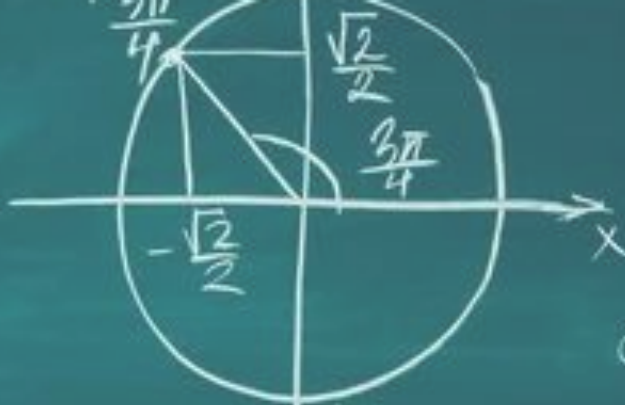
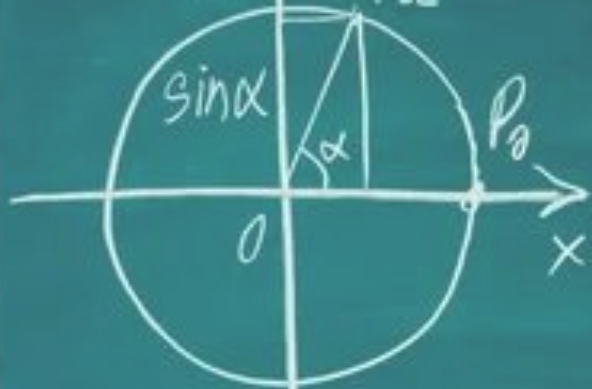




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = -\sin, \quad E(\sin) = \cos, \quad E(\cos) = -\sin, \quad [1; 1]$$

First year Aerospace Engineering

Differential Calculus Exercises

Marcos Carrera Valle

Find the derivative of the following functions

- 1) $f(x) = 4x^3 + 2e^{(3x-1)}$
- 2) $f(x) = 3x^{(3/4)} - \ln(5x)$
- 3) $f(x) = 2x^{(-4)} + 5\sin(3x)$
- 4) $f(x) = 3\tan(2x) - 4e^{(-x)}$
- 5) $f(x) = 6x^{(5/2)} + 2\cos(4x)$
- 6) $f(x) = 3x^{(-2)} + 4e^{(2x)} - \ln(x^3)$
- 7) $f(x) = 5x^{(1/2)} - 2\sin(0.5x) + x$
- 8) $f(x) = 7e^{(0.5x)} + 3x^{(-1/3)}$
- 9) $f(x) = \ln(4x) + 2\tan(x) - 5x^2$
- 10) $f(x) = 2x^5 - 3e^{(2x)} + 4\cos(3x)$

Find the derivative of the following functions

- 11) $f(x) = 5^{(2x)} + 4x^{1.6}$
- 12) $f(x) = 3e^{(0.4x+1.2)} - 2\cos(5x)$
- 13) $f(x) = 0.5x^{(-2.4)} + \ln(3x^2)$
- 14) $f(x) = 2\tan(0.5x) - 4^{(0.5x)}$
- 15) $f(x) = 6x^{(0.75)} + 3\sin(2x) - e^{(-1.5x)}$
- 16) $f(x) = 7(0.2)^x + 2x^{(-0.5)}$
- 17) $f(x) = \ln(0.1x) + 3\cos(4x) - 2x^{2.2}$
- 18) $f(x) = 4e^{(2.5x-3)} + 5\tan(3x)$
- 19) $f(x) = 3^x - 2x^{(-1.2)} + 4x$
- 20) $f(x) = 0.6x^5 + 2(1.5)^x - \ln(x^{-3})$

Find the derivative of the following functions

21) $f(x) = x^2 * e^{(3x)}$

22) $f(x) = (2x + 1)^5$

23) $f(x) = \sin(x) / x^2$

24) $f(x) = \ln(x^2 + 1)$

25) $f(x) = (3x + 4) / (x - 2)$

26) $f(x) = e^{(2x)} * \cos(3x)$

27) $f(x) = (x^2 + 1)^{(1/2)}$

28) $f(x) = x * \ln(x)$

29) $f(x) = \tan(x) / e^x$

30) $f(x) = \cos^3(x)$

Find the derivative of the following functions

31) $f(x) = (x + 1)^2 * e^{(-x)}$

32) $f(x) = 5^{(x^2)}$

33) $f(x) = x^{(1/2)} * \sin(2x)$

34) $f(x) = e^{(2x)} / (x + 1)$

35) $f(x) = \sin(\cos(x))$

36) $f(x) = x^2 / (3x + 1)$

37) $f(x) = (x^2 + 1) * \ln(x)$

38) $f(x) = \ln(x) / x^3$

39) $f(x) = (e^x + 1) / (e^x - 1)$

40) $f(x) = \sin(x) * \cos(2x)$

Find the local maxima/ minima of the following functions. Additionally, find the global maxima/ minima:

41) $f(x) = x^2 e^{-x}$ on the interval $[-1, 4]$

42) $f(x) = x / (x^2 + 1)$ on the interval $[-2, 3]$

43) $f(x) = \sin^2(x) - \cos(x)$ on the interval $[0, 2\pi]$

44) $f(x) = x(x - 3)^3$ on the interval $[0, 4]$

45) $f(x) = (x^2 - 1)^2$ on the interval $[-2, 2]$

You are an aeronautical engineer designing a flight path for a new high-altitude solar drone.

The drone's "Endurance Index" E , which measures how efficiently it uses solar energy relative to its speed, is dependent on the velocity in knots.

The engineering team has determined that the Endurance Index is modeled by the following function: $E(v) = v^2 * e^{0.01v}$ where $0 \leq v \leq 300$.

46) Find the optimal velocity (v) that maximizes the Endurance Index.

You are testing a new wing profile for a supersonic jet. The performance of the wing depends on the "Angle of Attack" (α), measured in degrees.

To determine the efficiency of the wing, you need to analyze the relationship between the Lift it generates and the Drag it creates.

Instrumentation has provided two separate functions for these forces:

Lift Coefficient: $L(\alpha) = 10\alpha^{0.8}$

$D(x) = 0.04 + 0.02\alpha^{2.8}$

47) Find the Angle of Attack (α) that maximizes the Aerodynamic Efficiency L/D .

You are designing the air inlet for a "Scramjet" (Supersonic Combustion Ramjet) fighter capable of Mach 6. The geometry of the inlet ramp is crucial. The performance is determined by the "Ramp Angle" (θ), measured in radians, where $0 < \theta < 1.5$.

Two competing physical phenomena have been modeled by the physics team:

Compression Efficiency: $C(\theta) = 10(\tan(\theta))^{0.5}$

(Higher angles create stronger shockwaves, compressing air better for combustion).

Thermal Drag Factor: $D(\theta) = e^{(2\theta)}$

(Higher angles create massive friction and heat, increasing drag exponentially).

48) Find the Ramp Angle that maximizes the Total Inlet Performance.

The Performance P is defined as the Compression Efficiency divided by the Thermal Drag Factor.

A military prototype Jump-Jet is performing a vertical takeoff and landing test. The telemetry team records the vertical altitude, $x(t)$ (in meters), as a function of time, t (in seconds), for the interval $0 \leq t \leq 10$.

The altitude is modeled by the function: $x(t) = 20 * t^3 * e^{(-0.6t)}$

49)

Find the time (t) when the Velocity is at its maximum. What is the maximum velocity?

Find the time (t) when the Acceleration is at its maximum (positive). What is $a_{\max}$?

You are designing the fuel injection system for a high-performance racing engine. You need to place two piezoelectric pressure sensors inside the fuel pipe to monitor the flow.

The pipe is not a simple cylinder; it narrows and widens to control pressure.

The Area of the pipe (A) in cm^2 varies with the Position (x) in cm upstream of the injector, modeled by: $A(x) = 0.2x^2 + 1$ (where $-3 \leq x \leq 3$).

The Velocity of the fuel (V) in m/s depends on the Area, modeled by: $V(A) = 100 / A$

The sensors must be placed where the fuel experiences the most extreme forces to prevent cavitation (bubbles forming). Force is proportional to Acceleration.

50) Find the positions (x) where the Acceleration of the fuel is at its Maximum (most positive) and Minimum (most negative).

SOLUTIONS

Find the derivative of the following functions

- 1) $f(x) = 12x^2 + 6e^{(3x-1)}$
- 2) $f(x) = (9/4)x^{(-1/4)} - 1/x$
- 3) $f(x) = -8x^{(-5)} + 15\cos(3x)$
- 4) $f(x) = 6\sec^2(2x) + 4e^{(-x)}$
- 5) $f(x) = 15x^{(3/2)} - 8\sin(4x)$
- 6) $f(x) = -6x^{(-3)} + 8e^{(2x)} - 3/x$
- 7) $f(x) = (5/2)x^{(-1/2)} - \cos(0.5x) + 1$
- 8) $f(x) = 3.5e^{(0.5x)} - x^{(-4/3)}$
- 9) $f(x) = 1/x + 2\sec^2(x) - 10x$
- 10) $f(x) = 10x^4 - 6e^{(2x)} - 12\sin(3x)$

Find the derivative of the following functions

- 11) $f(x) = 2(5)^{(2x)}\ln(5) + 6.4x^{0.6}$
- 12) $f(x) = 1.2e^{(0.4x+1.2)} + 10\sin(5x)$
- 13) $f(x) = -1.2x^{(-3.4)} + 2/x$
- 14) $f(x) = \sec^2(0.5x) - 0.5(4)^{(0.5x)}\ln(4)$
- 15) $f(x) = 4.5x^{(-0.25)} + 6\cos(2x) + 1.5e^{(-1.5x)}$
- 16) $f(x) = 7(0.2)^x\ln(0.2) - x^{(-1.5)}$
- 17) $f(x) = 1/x - 12\sin(4x) - 4.4x^{1.2}$
- 18) $f(x) = 10e^{(2.5x-3)} + 15\sec^2(3x)$
- 19) $f(x) = 3^x\ln(3) + 2.4x^{(-2.2)} + 4$
- 20) $f(x) = 3x^4 + 2(1.5)^x\ln(1.5) + 3/x$

Find the derivative of the following functions

21) $f(x) = 2xe^{(3x)} + 3x^2e^{(3x)}$

22) $f(x) = 10(2x + 1)^4$

23) $f(x) = (x \cos(x) - 2 \sin(x)) / x^3$

24) $f(x) = 2x / (x^2 + 1)$

25) $f(x) = -10 / (x - 2)^2$

26) $f(x) = e^{(2x)} * (2 \cos(3x) - 3 \sin(3x))$

27) $f(x) = x / (x^2 + 1)^{(1/2)}$

28) $f(x) = \ln(x) + 1$

29) $f(x) = (\sec^2(x) - \tan(x)) / e^x$

30) $f(x) = -3 \cos^2(x) \sin(x)$

Find the derivative of the following functions

31) $f(x) = (x + 1)e^{(-x)} * (1 - x)$

32) $f(x) = 2x * 5^{(x^2)} * \ln(5)$

33) $f(x) = 0.5x^{(-1/2)}\sin(2x) + 2x^{(1/2)}\cos(2x)$

34) $f(x) = e^{(2x)}(2x + 1) / (x + 1)^2$

35) $f(x) = -\sin(x)\cos(\cos(x))$

36) $f(x) = (3x^2 + 2x) / (3x + 1)^2$

37) $f(x) = 2x\ln(x) + (x^2 + 1)/x$

38) $f(x) = (1 - 3\ln(x)) / x^4$

39) $f(x) = -2e^x / (e^x - 1)^2$

40) $f(x) = \cos(x)\cos(2x) - 2\sin(x)\sin(2x)$

Find the local maxima/ minima of the following functions. Additionally, find the global maxima/ minima:

- 41) $f(x) = x^2 e^{-x}$ on the interval $[-1, 4]$
Global Max: $(-1, e)$, Global Min: $(0, 0)$, Local Max: $(2, 4e^{-2})$
- 42) $f(x) = x / (x^2 + 1)$ on the interval $[-2, 3]$
Global Max: $(1, 0.5)$, Global Min: $(-1, -0.5)$
- 43) $f(x) = \sin^2(x) - \cos(x)$ on the interval $[0, 2\pi]$
Global Max: 1.25 (at $x = 2\pi/3$ and $4\pi/3$), Global Min: -1 (at $x = 0$ and 2π), Local Min: $(\pi, 1)$
- 44) $f(x) = x(x - 3)^3$ on the interval $[0, 4]$
Global Max: $(4, 4)$, Global Min: $(0.75, -8.54)$, Local Max: $(3, 0)$ is a stationary point of inflection, not a strict max/min, but $f(0)$ is a local max relative to the interval start.
- 45) $f(x) = (x^2 - 1)^2$ on the interval $[-2, 2]$
Global Max: 9 (at $x = -2$ and 2), Global Min: 0 (at $x = -1$ and 1), Local Max: $(0, 1)$

46) The optimal velocity to maximize the Endurance Index is 200 knots.

47) $\alpha = 0.923$ deg

48) $\theta = 15$ deg

49) $V_{\max} =$ at $t = 2.11$ sec, $a_{\max} =$ at $t = 0.75$ sec

50) $x = -1$ (Maximum acceleration), $x = 1$ (Minimum acceleration)